Eine Woche, ein Beispiel

4.12. hyperbolicity notations

This document records notations in hyperbolicity taken during the Chemnitz conference.

$$\begin{tikzcd}
 & & & Y-X \text{ is Brhyp} \arrow[dll, dashed] \arrow[dl, dashed] \\
\text{HE in } Y \arrow[r, Rightarrow] & \text{Khyp} \arrow[r, Rightarrow, "B(p_1 X - \{p_1, p_2, p_3\})"'] & \text{Bohyp} \arrow[r, Rightarrow] & \text{Brhyp} \\
 & & \text{Phyp} \arrow[ur, Rightarrow] &
\end{tikzcd}$$

X proj: all hyperbolicity conditions are the same except HE in Y

Moreover, when $X \subseteq A$ is contained in AV,
hyp $\Leftrightarrow$ X don't contain a translation of an pos-dim AV.

Schwarz's lemma $\mathbb{D} := \{z \in \mathbb{C} \mid |z| < 1\}$

$f: \mathbb{D} \longrightarrow \mathbb{D}$ holo, then $\forall\, p, q \in \mathbb{D}$,

$$d_{\mathbb{D}}(f(p), f(q)) \leq d_{\mathbb{D}}(p, q)$$

$\rightsquigarrow$ pseudodistance d_X.

Def. X is Kobayashi hyperbolic (Khyp), if

$$d_X(x, y) = 0 \Rightarrow x = y.$$

Fact. (1) X, Y is Khyp $\Rightarrow$ $X \times Y$ is Khyp

(2) For a cover $\tilde{X} \longrightarrow X$, $\tilde{X}$ is Khyp $\Leftrightarrow$ X is Khyp

(3) For a RS X, X is Khyp $\Leftrightarrow$ $\tilde{X}^{univ} \cong \mathbb{D}$

(4) For a holo map $f: X \longrightarrow Y$,

$$d_Y(f(p), f(q)) \leq d_X(p, q).$$

Def. X quasi-proj. X is Borel hyperbolic (Bohyp), if $\forall$ qproj Y, $\forall f: Y \longrightarrow X$ holo, f is alg.

Def X is Brody hyperbolic (Brhyp), if $\forall f: \mathbb{C} \longrightarrow X$ hol, f is constant.

Bohyp $\Rightarrow$ Brhyp: if f is holo, then f & $f \circ \exp$ are all algebraic, which is only possible when f is constant.

Def X qproj. $\overline{X}$ is a projective compactification of X.
X is Picard hyperbolic (Phyp) if

$$\begin{tikzcd} \forall\ \mathbb{D}^* \arrow[r, "f"] \arrow[d, hook] & X \text{ holo} \arrow[d, hook] \\ \mathbb{D} \arrow[r, dashed, "\exists \bar{f}"] & \overline{X} \end{tikzcd}$$

Def For X cplx mfld, $\xi \in T_{x_0}X$, define

$$F_X(\xi) := \inf \left\{ \lambda \in \mathbb{R}_{>0} \;\middle|\; \begin{array}{l} \exists f: \mathbb{D} \longrightarrow X \text{ holo}, \\ f(0) = x_0,\ f'(0) = \lambda^{-1}\xi \end{array} \right\}$$

For $\mathcal{U} \subset X$ open, $\mathcal{U}$ is hyperbolically embedded in X (HE in X), if $\forall$ metric h on TX, $\exists\, \varepsilon > 0$ s.t.

$$\varepsilon \cdot \|\cdot\|_h \big|_{\mathcal{U}} \leq F_{\mathcal{U}}.$$